USER GUIDE - AuraFlex Gym Management System

Group Number : 11

End-User Manual

Version: 1.0

Date: 31 October 2026

1. About AuraFlex

AuraFlex is a gym management program that runs in the terminal. It helps the gym staff to manage members, memberships, payments, attendance, trainers, trainer assignments, and reports. The program saves all its records in text files inside the `data/` folder.

This manual is for administrators, receptionists, and trainers who use the program. It does not explain how to change the source code or how to edit the data files directly.

2. Compiling the Application (Ubuntu / Linux)

The program must be compiled before it can be used. Follow these steps on Ubuntu or Linux. A C compiler (GCC) is needed.

1. Open a terminal in the AuraFlex project folder.
2. Create the `build` folder where the compiled program will be saved:

```
mkdir -p build
```

3. Compile all the source files with this command:

```
gcc -Wall -Wextra -std=c11 -D_POSIX_C_SOURCE=200809L -Iinclude \
  src/main.c \
  src/auth/auth.c \
  src/member/member.c \
  src/membership/membership.c \
  src/trainers/trainer.c \
  src/payment/payment.c \
  src/attendance/attendance.c \
  src/reports/reports.c \
  src/system/system.c \
  src/system/ui.c \
  -o build/gym
```

4. If there are no errors, the program is created as `build/gym`.

A short explanation of the command:

- `-Wall -Wextra` show extra warning messages while compiling.
- `-std=c11` compiles the code using the C11 standard.
- `-D_POSIX_C_SOURCE=200809L` turns on the POSIX features that the program uses.
- `-Iinclude` tells the compiler to look for the header files in the `include` folder.
- `-o build/gym` gives the name and the place of the finished program.

You only need to compile again if the source code is changed.

3. Starting the Application

1. Open a terminal in the AuraFlex project folder.
2. Make sure the `data/` folder is there.
3. Run the program with this command:

```
./build/gym
```

4. Wait until the login screen appears.

The program uses paths such as `data/users.txt`, so it must be started from the project folder. If it is started from another folder, it may not find the files.

4. Signing In

1. Type your username when the program asks for it.
   admin ----- 123
   Trainer ----- 123
   reception ---- 123
2. Type your password.
3. The password is hidden while you type, if the system supports it.
4. If the username and password are correct and the user is active, your dashboard opens.

You can try to log in a maximum of three times. Wrong details and inactive users are rejected. After three failed attempts, the program shows an access denied message and closes. If you cannot sign in, contact the system administrator. Do not edit `data/users.txt` by yourself.

5. Dashboard Access

Administrator

The administrator dashboard has these options:

1. Member Management
2. Membership Plans / Membership Management
3. Trainer Management

4. Payment Management
5. Attendance Management
6. Reports
7. Logout

Receptionist

The receptionist dashboard has these options:

1. Register Member
2. Manage Membership
3. Record Payment
4. Record Attendance
5. View Members
6. Logout

Trainer

The trainer dashboard has these options:

1. View Assigned Members
2. Assign Member
3. Remove Member
4. Logout

The names of the menu options may be a little different in some versions, but the operations and the permissions of each role are the same.

6. Member Management

Member management can be used by administrators. Receptionists can use some of the member operations.

Register a member

1. Open **Member Management** or **Register Member**.
2. Enter a new member ID.
3. Enter the name, phone number, and email address of the member.
4. Confirm the details when the program asks.
5. The member is saved with the status `ACTIVE`.

Every member ID must be different. Do not use an ID that is already used.

View or search members

- Choose **View Members** to see the list of all members.
- Choose **Search Member** and enter a member ID to find one member.
- Check the status of the member before you record a payment or attendance.

Update member details

1. Choose **Update Member**.
2. Enter the member ID.
3. Enter the new details.
4. Confirm the update.

Freeze or unfreeze a member

Use **Freeze Member** to change an active member to `FROZEN` for some time. Use **Unfreeze Member** to change the member back to `ACTIVE`.

A frozen member cannot make a payment or check in. The old records of the member are not deleted when the member is frozen.

7. Membership Management

1. Open **Membership Management**.
2. Choose **Assign Membership**.
3. Enter a new membership ID and the ID of an existing member.
4. Enter the plan name, start date, end date, and fee.
5. Enter or confirm the payment details when the program asks.
6. Confirm the operation.

The member must already exist. The dates must be valid, and the end date must not be earlier than the start date. Assigning a membership is connected to the payment process, so if the payment fails, the membership is not shown as successful.

Use **View Memberships** to see the saved memberships. Use the reports menu to find active memberships whose end date has already passed.

8. Payment Management

Record a payment

1. Open **Payment Management**.
2. Choose **Record Payment**.
3. Enter the member ID.
4. Enter a positive payment amount.
5. Enter a valid payment date.
6. Select one payment method: `CASH`, `CARD`, or `ONLINE`.
7. Confirm the payment.

The member must exist and must be active. The program creates the payment ID automatically from the existing payment records.

Check payments

- **Search Payment History:** finds the payments of one member.

- **View All Payments:** shows all the recorded payments.
- **Calculate Total Payments:** calculates the total amount saved in the payment file.

9. Attendance Management

Record check-in

1. Open **Attendance Management**.
2. Choose **Record Check-In**.
3. Enter the member ID.
4. Make sure the member is active.
5. The program saves the current date and time and creates an attendance ID.

A member cannot check in if the member does not exist, is frozen, or already has an open check-in on the same day.

Record check-out

1. Choose **Record Check-Out**.
2. Enter the member ID.
3. The program finds the open attendance record of the member and saves the current check-out time.

A check-out cannot be recorded if there is no open check-in.

Check attendance

- **Search Attendance:** searches the records using a member ID.
- **View Attendance History:** shows all the saved attendance records.
- **Daily Attendance Summary:** shows the check-ins, the completed check-outs, and the members who are still inside.

10. Trainer Management

Trainer management can be used by administrators.

Register or update a trainer

1. Open **Trainer Management**.
2. Choose **Register Trainer**.
3. Enter a new trainer ID and the details of the trainer.
4. Save the record.

Use **View Trainers**, **Search Trainer**, or **Update Trainer** to see or change the trainer records. Use **Change Trainer Status** to activate or deactivate a trainer.

Assign a member to a trainer

1. Choose **Assign Member to Trainer**.
2. Enter the trainer ID.

3. Enter the member ID.
4. Confirm the assignment.

The trainer must exist and be active, the member must exist, and the same assignment must not already be saved.

Remove or view assignments

- Choose **View Trainer Members** to see the assignments.
- Choose **Remove Member From Trainer** and enter the IDs to remove an assignment.

Removing an assignment does not delete the trainer or the member.

11. Reports

Reports can be used by administrators.

- **Member Report:** shows the member records.
- **Membership Report:** shows the membership records.
- **Expired Membership Report:** shows the active memberships that have expired.
- **Payment Report:** shows the payment details and the total.
- **Summary Report:** shows the important counts and totals of the system together.

The reports use the files saved in the `data/` folder. If a record you just entered is not shown, check that the last operation showed a success message.

12. Input Rules

Please follow these rules when you enter information:

- Use different IDs for members, trainers, memberships, and assignments.
- Enter only numbers when choosing a menu option.
- Enter positive amounts for payments.
- Enter valid dates in the format that the program asks for.
- Use only `CASH`, `CARD`, or `ONLINE` as the payment method.
- Use the ID of an existing member for memberships, payments, attendance, and trainer assignments.
- Do not use spaces in the fields that are saved in the text files. Use an underscore (for example `John_Silva`) when more than one word is needed.
- Do not close the terminal while a record is being saved.

If the input is wrong, the program shows a message and lets you try again. An operation that failed should not be treated as completed.

13. Returning to Menus and Logging Out

After an operation, follow the message on the screen to press Enter or go back to the menu. Use the **Return** option to leave a submenu. Choose **Logout** on the dashboard when you finish.

Always log out before closing the terminal, especially on a shared computer.

14. Troubleshooting

Problem	What to do
Login is rejected	Check the username, password, spelling, and account status. Contact an administrator after three failed attempts.
Compiling fails or `build/gym` is missing	Make sure you are in the project folder, the `build` folder was created, and all the source files are in the gcc command.
A file cannot be opened	Make sure the program was started from the project folder and the `data/` folder is there.
Member ID is rejected	Check that the member exists, is active where needed, and that the ID is typed exactly.
Duplicate ID message	Enter a new ID that is not already saved.
Payment is rejected	Use an active member, a positive amount, a valid date, and `CASH`, `CARD`, or `ONLINE`.
Check-in is rejected	Check that the member is active and does not already have an open check-in today.
Check-out is rejected	Check that the member has checked in and has not already checked out.
Assignment is rejected	Check that the trainer is active, the member exists, and the assignment is not already saved.
New data is missing	Go back to the menu after the success message and open the view or report again.
Screen looks wrong	Resize the terminal or use a command-line program that works properly.

Do not change or delete the data files to fix an error. Contact the system maintainer, because changes can break the record formats and the links between the components.

15. Data and Security Notes

- The records are saved as plain text and are not encrypted.
- Passwords are saved in the user data file, as in the project design.
- Allow only the right people to open the project folder and the `data/` folder.
- Keep a backup of the `data/` folder before maintenance or testing.
- Do not share passwords or copy real member information into test files.

16. Quick Reference

```
Compile the program (first time only)
  -> Run ./build/gym
  -> Sign in
  -> Select the role dashboard
  -> Choose a component
  -> Enter and check the details
  -> Confirm success
  -> Return to the dashboard
  -> Log out
```

For technical details, the source layout, and the implementation, see `docs/FINAL_PROJECT_REPORT.md`.

For the formal system requirements, see `docs/SRS.md`.